

Roll No.

TME-503

B. TECH. (ME) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

DYNAMICS OF MACHINES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the conditions for a body to be in equilibrium under the action of two forces, three forces and two forces and a torque ?

10 Marks (CO1)

OR

- (b) Determine the required input torque on the slider-crank mechanism for the static equilibrium when the applied piston load is 1500 N. The lengths of crank and connecting rod are 40 mm and 100 mm respectively and crank has turned through 45° from the inner-dead centre.

10 Marks (CO1)

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2. (a) What is the principle of virtual work ? Explain. 10 Marks (CO1, CO2)

OR

- (b) For a certain engine having an average speed of 1200 r.p.m., a flywheel approximated as a solid disc, is required for keeping the fluctuation of speed within 2% about the average speed. The fluctuation of kinetic energy per cycle is found to be 2 kJ. What is the least possible mass of the flywheel if its diameter is not to exceed 1 m ?

10 Marks (CO1, CO2)

3. (a) Define free body diagram. Draw free body diagram to illustrate static force analysis of four bar mechanism and single slider chain mechanism.

10 Marks (CO1, CO2)

OR

- (b) The turning-moment diagram for a multicylinder engine has been drawn to a vertical scale of 1 mm = 650 N-m and a horizontal scale of 1 mm = 45° . The areas above and below the mean torque line are - 28, + 380, - 260, + 310, - 300, + 242, - 380, + 265 and - 229 mm².

The fluctuation of speed is limited to $\pm 1.8\%$ of the mean speed which is 400 r.p.m. The density of the rim material is 7000 kg/m³ and width of the rim is 4.5 times its thickness. The centrifugal stress (hoop stress) in the rim material is limited to 6 N/mm². Neglecting the effect of the boss and arms, determine the diameter and cross-section of the flywheel rim.

10 Marks (CO1, CO2)

4. (a) Derive an expression for turning moment at crankshaft in terms of piston effort and the angle turned by the crank. 10 Marks (CO2)

(3)

OR

- (b) The turning moment diagram for a petrol engine is drawn to a vertical scale of 1 mm to 6 N-m and a horizontal scale on 1 mm to 1 degree. The turning moment repeats itself after every half revolution of engine. The area above and below the mean torque line are 305, 710, 50, 350, 980 and 275 mm². The rotating parts amount to a mass of 40 kg at a radius of gyration of 140 mm. Calculate the coefficient of fluctuation of speed if the speed of the engine is 1500 r.p.m. 10 Marks (CO2)

5. (a) Derive an expression for the velocity and acceleration of reciprocating part of slider crank mechanism. 10 Marks (CO2)

OR

- (b) In a vertical steam engine, the connecting rod is 4.5 times the crank. The weight of the reciprocating parts is 120 kg and the stroke of the piston is 440 mm. The engine runs at 250 r.p.m. If the net load on the piston due to steam pressure is 25 kN when the crank has turned through an angle of 120° from the top dead centre, determine :

- (i) Piston effort
- (ii) Crank effort
- (iii) Turning moment on the crankshaft

10 Marks (CO2)